



Roma Tre



DIPARTIMENTO
DI MATEMATICA
E FISICA



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Exactness of Ohm's law in extended quantum Hall systems

Speaker: Domenico Monaco - Sapienza Università di Roma

Time: Wednesday October 7, 13:00-14:00

Location: Room B308, Largo S. Murialdo 1 (Roma)

In the quantum Hall effect, when a 2D magnetic electron gas is brought out of equilibrium by an electric field of small intensity, the transverse current responds exhibiting a quantized conductivity. This linear response coefficient can be expressed through the so-called double commutator formula, involving only the equilibrium ground state of the system, and furthermore all higher-order terms in this response vanish, establishing exactness of Ohm's law. I will show how these results can be proved by means of a recently developed technique which characterizes a non-equilibrium almost-stationary state of the perturbed quantum system. This technique applies to continuum models of quantum Hall systems in the 1-body approximation, as well as to weakly interacting Hofstadter-like models for lattice fermions. (Based on joint works with G. Marcelli, T. Miyao, S. Teufel, and M. Wesle.)